

Exp No: 01

Transformation between Cartesian, Cylindrical and spherical coordinate system

Aim:

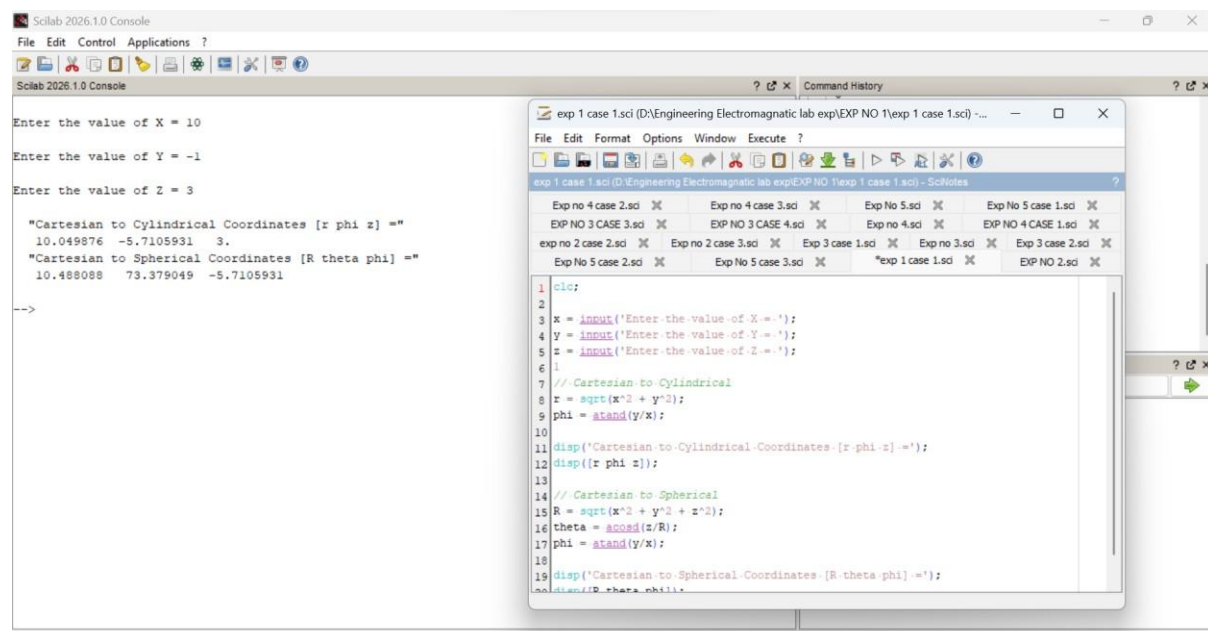
To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

Test case (i) Transformation of Cartesian coordinate points into Cylindrical and spherical coordinate system



The screenshot displays the SCILAB 2026.1.0 interface. The main console window shows the execution of a script. The user is prompted to enter values for X, Y, and Z. The script then calculates and displays the Cartesian to Cylindrical Coordinates [r phi z] and Cartesian to Spherical Coordinates [R theta phi].

```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console
Enter the value of X = 10
Enter the value of Y = -1
Enter the value of Z = 3
"Cartesian to Cylindrical Coordinates [r phi z] ="
10.049876 -5.7105931 3.
"Cartesian to Spherical Coordinates [R theta phi] ="
10.488088 73.379049 -5.7105931
-->
```

The Command History window shows the following code:

```
1 clc;
2
3 x = input('Enter the value of X = ');
4 y = input('Enter the value of Y = ');
5 z = input('Enter the value of Z = ');
6
7 // Cartesian to Cylindrical
8 r = sqrt(x^2 + y^2);
9 phi = atan2(y/x);
10
11 disp('Cartesian to Cylindrical Coordinates [r phi z] =');
12 disp([r phi z]);
13
14 // Cartesian to Spherical
15 R = sqrt(x^2 + y^2 + z^2);
16 theta = acos(z/R);
17 phi = atan2(y/x);
18
19 disp('Cartesian to Spherical Coordinates [R theta phi] =');
20 disp([R theta phi]);
```

Practical Calculation:

EXP-1

Transformation between cartesian, cylindrical and spherical system

Case (i)

$$x = 10$$

$$y = -1$$

$$z = 3$$

Cartesian to cylindrical coordinates

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$z = z$$

$$r = \sqrt{10^2 + (-1)^2} = \sqrt{100 + 1} = \sqrt{101} = 10.05$$

$$\phi = \tan^{-1} \left(\frac{-1}{10} \right) = \tan^{-1} (-0.1) = -5.71$$

$$\phi = -5.71$$

$$z = 3$$

$$r = 10.05$$

$$\phi = -5.71$$

$$z = 3$$

Cartesian to spherical coordinates

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \left(\frac{z}{R} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$R = \sqrt{10^2 + (-1)^2 + 3^2} = \sqrt{110} = 10.49$$

$$\theta = \cos^{-1} \left(\frac{3}{10.49} \right) = (0.286) \quad \theta = 73.32$$

$$\phi = \tan^{-1} \left(\frac{-1}{10} \right)$$

$$= -5.71$$

$$R = 10.49$$

$$\theta = 73.32$$

$$\phi = -5.71$$

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

The screenshot shows the Scilab 2026.1.0 Console and Command History. The console displays the input values and the resulting coordinates. The Command History window shows the script being executed.

```

Scilab 2026.1.0 Console
File Edit Control Applications ?
Enter the value of r = 4
Enter the value of phi (in degrees) = 35
Enter the value of z = -5

"Cylindrical to Cartesian Coordinates [x y z] ="
3.276082 2.2943057 -5.
"Cylindrical to Spherical Coordinates [R theta phi] ="
6.4031242 141.34019 35.
-->

exp 1 case 2.sci (D:\Engineering Electromagnetic lab exp\EXP NO 1\exp 1 case 2.sci) - ...
File Edit Format Options Window Execute ?
exp 1 case 2.sci (D:\Engineering Electromagnetic lab exp\EXP NO 1\exp 1 case 2.sci) - SciNotes
Exp no 4 case 2.sci Exp no 4 case 3.sci Exp No 5.sci Exp No 5 case 1.sci Exp No 5 case 2.sci
EXP NO 3 CASE 3.sci EXP NO 3 CASE 4.sci Exp no 4.sci Exp no 4 CASE 1.sci Exp NO 4 CASE 1.sci
exp no 2 case 2.sci Exp no 2 case 3.sci Exp 3 case 1.sci Exp no 3.sci Exp 3 case 2.sci
Exp No 5 case 3.sci *exp 1 case 1.sci EXP NO 2.sci exp 1 case 2.sci

1 clear;
2 clear;
3
4 // Input values
5 r = input('Enter the value of r = ');
6 phi = input('Enter the value of phi (in degrees) = ');
7 z = input('Enter the value of z = ');
8
9 // Cylindrical to Cartesian
10 x = r * cosd(phi);
11 y = r * sind(phi);
12
13 disp('Cylindrical to Cartesian Coordinates [x y z] =');
14 disp([x y z]);
15
16 // Cylindrical to Spherical
17 R = sqrt(r^2 + z^2);
18 theta = acosd(z / R);
19
20 disp('Cylindrical to Spherical Coordinates [R theta phi] =');
  
```

Practical Calculation:

Case (ii)

$r = 4$ $\phi = 35^\circ$ $z = -5$

Cylindrical to Cartesian (Coor)ing

$$x = r \cos \phi \quad y = r \sin \phi \quad z = z$$

$$x = 4 \cos 35^\circ \quad y = 4 \sin 35^\circ \quad z = -5$$

$$= 4(0.8132) \quad = 4(0.5736) \quad \boxed{z = -5}$$

$$\boxed{x = 3.2766} \quad \boxed{y = 2.2943}$$

Cylindrical to Spherical Coordinate

$$R = \sqrt{r^2 + z^2}$$

$$R = \sqrt{4^2 + (-5)^2}$$

$$= \sqrt{16 + 25}$$

$$= \sqrt{41}$$

$$\boxed{R = 6.4031}$$

$$\theta = \cos^{-1} \left(\frac{z}{R} \right)$$

$$\cos^{-1} \left(\frac{-5}{6.4031} \right)$$

$$\boxed{\theta = 141.34^\circ}$$

$$\phi = 35^\circ$$

$$\boxed{\phi = 35^\circ}$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

The screenshot shows the Scilab 2026.1.0 Console and Command History. The console displays the input values and the resulting coordinates.

```

Enter the value of r = 4
Enter the value of phi (in degrees) = 100
Enter the value of theta (in degrees) = 45

"Spherical to Cartesian Coordinates [x y z] ="
-0.4911512  2.785457  2.8284271
"Spherical to Cylindrical Coordinates [rho phi z] ="
2.8284271  100.  2.8284271
-->
  
```

The Command History window shows the script being executed:

```

1 //clear;
2 clear;
3 // Input
4 r = input('Enter the value of r = ');
5 phi = input('Enter the value of phi (in-degrees) = ');
6 theta = input('Enter the value of theta (in-degrees) = ');
7 // Spherical to Cartesian
8 x = r * sind(theta) * cosd(phi);
9 y = r * sind(theta) * sind(phi);
10 z = r * cosd(theta);
11 disp('Spherical to Cartesian Coordinates [x y z] =');
12 disp([x y z]);
13 // Spherical to Cylindrical
14 rho = r * sind(theta);
15 phi_cyl = phi;
16 z_cyl = z;
17 disp('Spherical to Cylindrical Coordinates [rho phi z] =');
18 disp([rho phi_cyl z_cyl]);
  
```

Practical Calculation:

Case (iii)

$r = 5$, $\theta = 45^\circ$, $\phi = 100^\circ$

Spherical to Cartesian:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Calculation:

$$x = 5 \sin 45^\circ \cos 100^\circ$$

$$= 5 \times 0.7071 \times (-0.1736)$$

$$x = -0.6139$$

$$y = 5 \sin 45^\circ \sin 100^\circ$$

$$= 5 \times 0.7071 \times 0.9848$$

$$y = 3.4818$$

$$z = 5 \cos 45^\circ$$

$$= 5 \times 0.7071$$

$$z = 3.5355$$

Spherical to Cylindrical:

$$\rho = r \sin \theta$$

$$\phi = \phi$$

$$z = r \cos \theta$$

Calculation:

$$\rho = 5 \sin 45^\circ$$

$$= 5 \times 0.7071$$

$$\rho = 3.5355$$

$$\phi = 100^\circ$$

$$z = 5 \cos 45^\circ$$

$$= 5 \times 0.7071$$

$$z = 3.5355$$

Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its classified two categories

(a) Cartesian to cylindrical and spherical coordinate system

(b) Cylindrical to cartesian and spherical coordinate system

(c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.

Parameters	Observed/Simulated Result	Theoretical Result
Case 1: x = 10, y = -1, z = 3	$\rho = 10.049876$, $\varphi = 354.289407^\circ$, z = 3 $r = 10.488088$, $\theta = 73.379049^\circ$, $\varphi = 354.289407^\circ$	$\rho = 10.050$, $\varphi = 354.289^\circ$, z = 3 $r = 10.488$, $\theta = 73.379^\circ$, $\varphi = 354.289^\circ$
Case 2: $\rho = 4$, $\varphi = 35^\circ$, z = -5	x = 3.276608, y = 2.294306, z = -5 $r = 6.403124$, $\varphi = 35^\circ$, $\theta = 141.340192^\circ$	x = 3.277, y = 2.294, z = -5 $r = 6.403$, $\varphi = 35^\circ$, $\theta = 141.340^\circ$
Case 3: r = 10, $\varphi = 100^\circ$, $\theta = 45^\circ$	x = -1.227878, y = 6.963642, z = 7.071068 $\rho = 7.071068$, $\varphi = 100^\circ$, z = 7.071068	x = -1.228, y = 6.964, z = 7.071 $\rho = 7.071$, $\varphi = 100^\circ$, z = 7.071